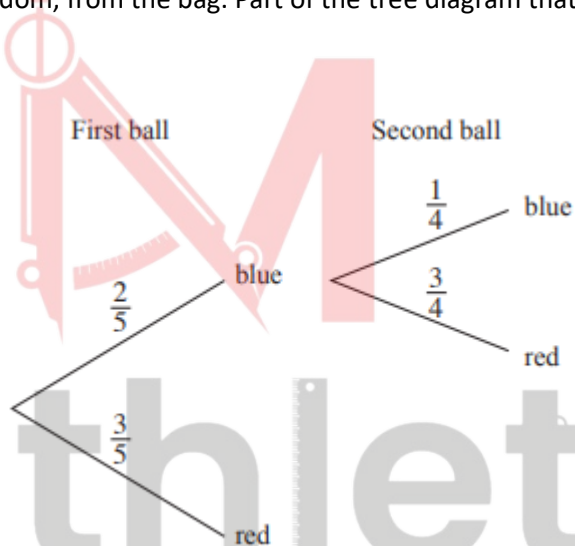


Probability

Worksheet 2 - With/Without Replacement (Non-Calculator)

1. W14/qp12/q21

A bag contains 5 balls, 2 of which are blue and 3 are red. One ball is taken, at random, from the bag. If it is red it is put back into the bag. If it is blue it is not put back into the bag. A second ball is taken, at random, from the bag. Part of the tree diagram that represents these outcomes is drawn below.

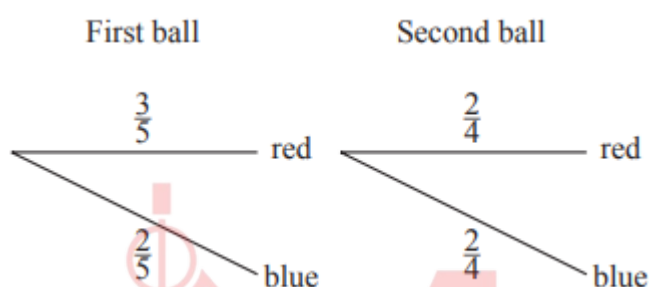


- Complete the tree diagram. [1]
- Expressing each answer as a fraction in its simplest form, find the probability that
 - both balls taken are blue, [1]
 - the second ball taken is blue. [2]

2. S15/qp11/q25

A bag contains 5 balls, of which 3 are red and 2 are blue. One ball is taken, at random, from the bag and is not replaced. If this ball is red, another ball is taken, at random, from the bag and is not replaced. This process is repeated until a blue ball is taken from the bag. Part of the tree diagram that represents these outcomes is drawn below.

- a. Complete the tree diagram. [2]



- b. Expressing each answer as a fraction, find the probability that
- the second ball taken is blue, [1]
 - a blue ball is the second, or the third, ball taken. [2]

3. W15/qp12/q23

A fair 4-sided spinner is numbered 1, 2, 3 and 4.

- a. Anil spins it once. He gets his score by doubling the number obtained. Complete the table to show the probabilities of his scores. [1]
- b. Billie spins it twice. She gets her score by adding the numbers obtained.
- Complete the possibility diagram. [1]

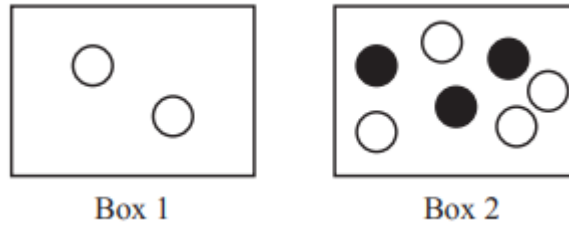
		First spin			
Second spin	+	1	2	3	4
	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4				

- ii. Complete the table showing the probabilities for some of Billie's scores. [1]

Score	> 2	> 4	> 6	> 8
Probability	$\frac{15}{16}$			

- c. Find the probability that Billie scores more than Anil. [2]

4. S16/qp11/q26



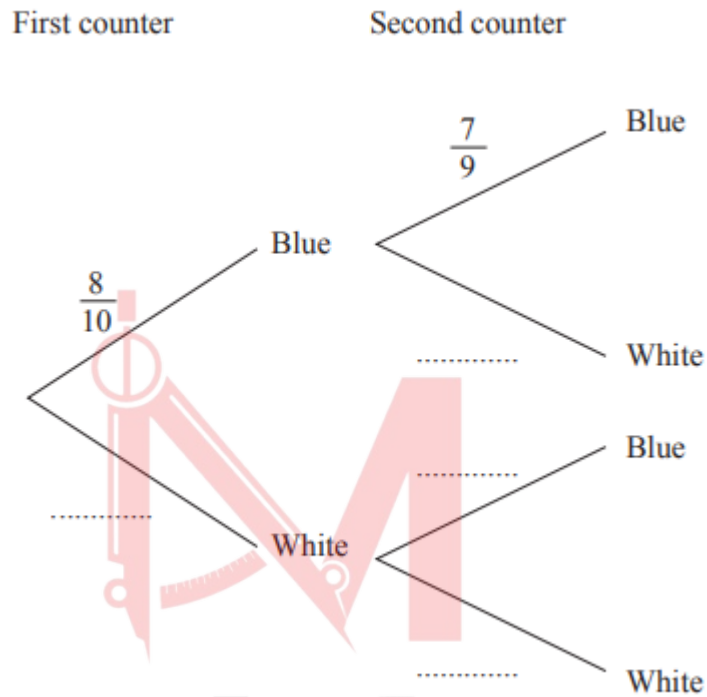
Box 1 contains 2 white balls. Box 2 contains 4 white balls and 3 black balls.

- a. Ann chooses, at random, one ball from each box.
- i. Find the probability that these balls are both black. [1]
 - ii. Find the probability that these balls have different colours. [1]
- b. From the original contents of Box 2, Belle chooses, at random, two balls without replacement. Find the probability that these balls are both white. [1]
- c. Carla chooses one of the boxes at random. With the original box contents, she then chooses, at random, one ball from this box. Find the probability that the ball is white. [2]

5. S16/qp12/q20

A bag contains 10 counters of which 8 are blue and 2 are white. Two counters are taken from the bag at random without replacement.

- a. Complete the tree diagram to show the possible outcomes and their probabilities. [1]



- b. Find, as a fraction, the probability that
- both counters are blue, [1]
 - one counter is blue and the other is white. [2]

= by Saad

6. W16/qp12/q18



Four cards are marked with the numbers 1, 2, 3 and 4. One card is chosen at random. A second card is then chosen, at random, from the remaining three cards. The sum of the numbers on the two chosen cards is calculated.

- a. Complete the table to show the possible outcomes. [1]

		First card			
Second card		1	2	3	4
	1				
	2				
	3				
	4				

- b. What is the probability that the sum is less than 2? [1]
 c. What is the probability that the sum is greater than 5? [1]

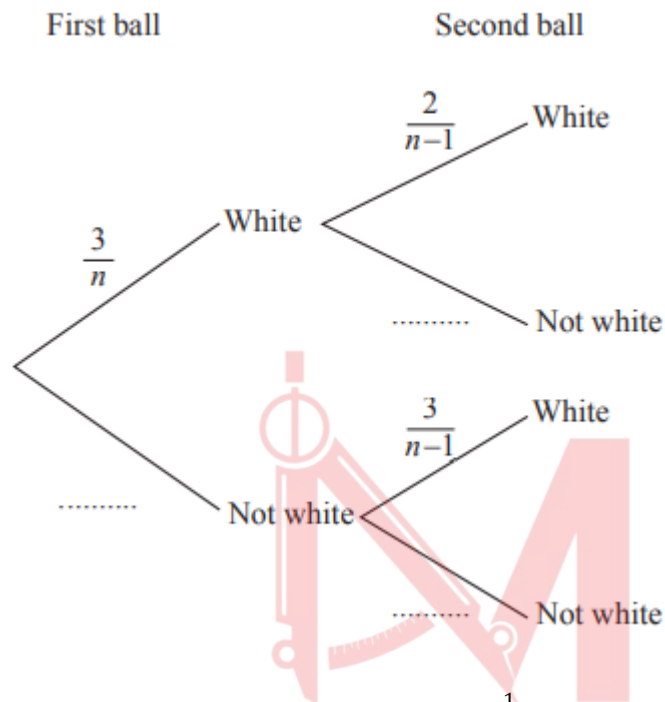
Mathlete[®]
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7. S17/qp12/q24

A bag contains n balls. 3 of the balls are white. Two balls are taken from the bag, at random, without replacement.

- a. Complete the tree diagram.

[2]



- b. The probability that both balls are white is $\frac{1}{15}$.

[2]

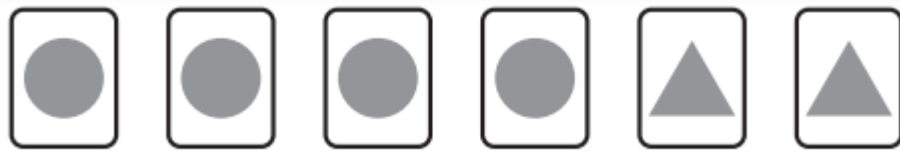
- c. Find the value of n .

[2]

8. S18/qp21/q20 (0580)

- a. A box contains 3 blue pens, 4 red pens and 8 green pens only. A pen is chosen at random from the box. Find the probability that this pen is green. [1]
- b. Another box contains 7 black pens and 8 orange pens only. Two pens are chosen at random from this box without replacement. Calculate the probability that at least one orange pen is chosen. [3]

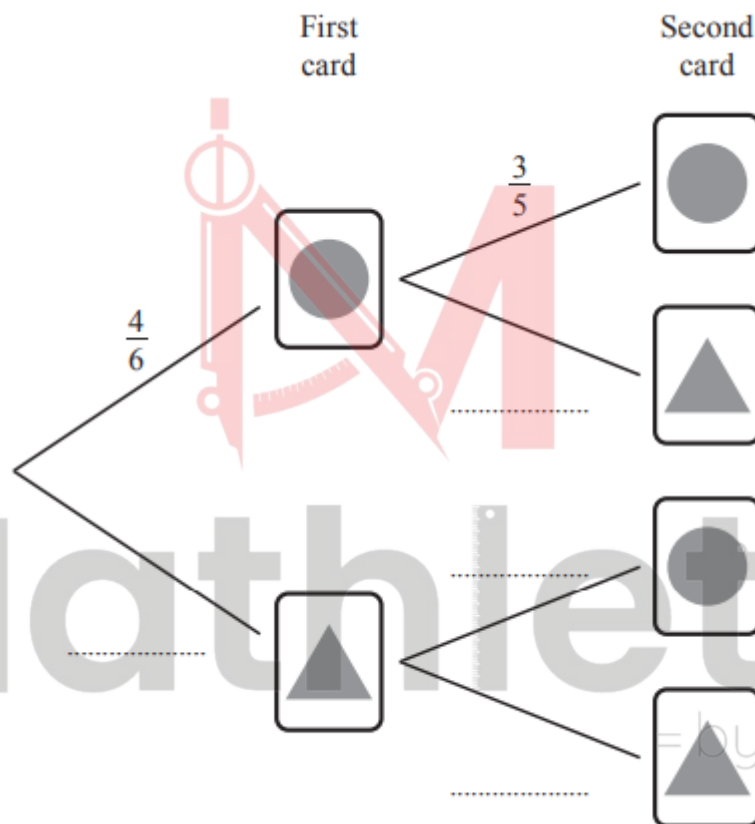
9. S19/qp12/q17



Nima has these six cards. Each card has a shape on it. She takes two cards at random without replacement.

a. Complete the tree diagram.

[2]



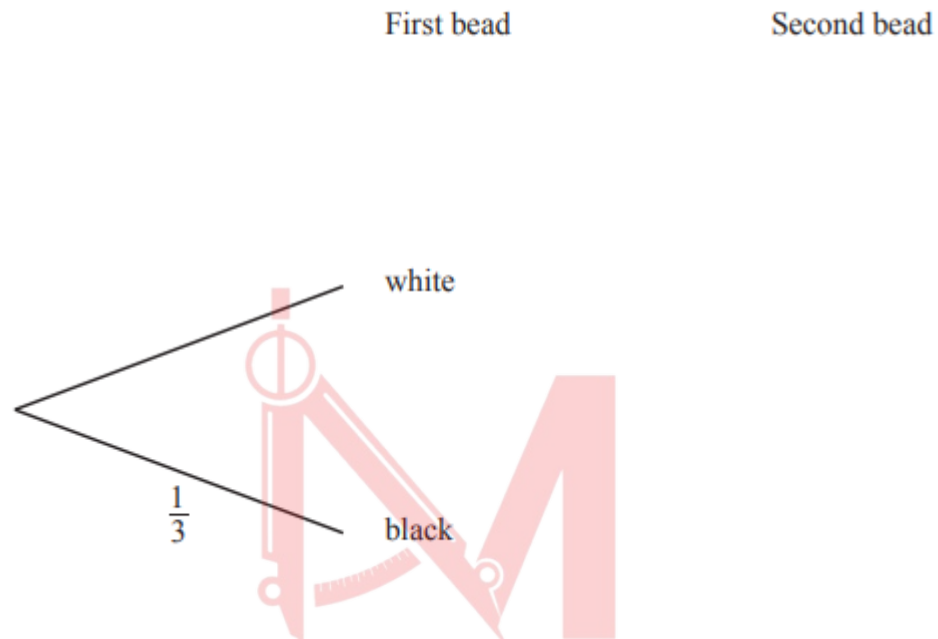
b. Find the probability that the shapes on Nima's two cards are the same. Give your answer as a fraction.

[2]

10. W19/qp12/q17

A bag contains two white beads and one black bead only. Two beads are taken, at random, without replacement from the bag.

- a. Complete the tree diagram. [2]



- b. Write down the probability that two black beads are taken. [1]

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= by Saad

Answer Key:

1. (a) Correct completion of tree diagram. (b)(i) $\frac{1}{10}$ (ii) $\frac{17}{50}$

2. (a) Correct third ball branches with $\frac{1}{3}$ and $\frac{2}{3}$ and correct fourth ball branch(es) (b) (i) $\frac{3}{10}$ (ii) $\frac{1}{2}$

3. (a) $\frac{1}{4} \frac{1}{4} \frac{1}{4} \frac{1}{4}$ (b) (i) 5 6 7 8 (ii) $\left(\frac{15}{16}\right) \frac{10}{16} \frac{3}{16}$ 0(c) $\frac{7}{16}$

4. (a) (i) 0 (ii) $\frac{3}{7}$ (b) $\frac{2}{7}$ (c) $\frac{11}{14}$

5. (a) $\frac{2}{10}, \frac{2}{9}, \frac{8}{9}, \frac{1}{9}$ (b) (i) $\frac{56}{90}$ oe (ii) $\frac{32}{90}$

6.(a)

–	3	4	5
3	–	5	6
4	5	–	7
5	6	7	–

 (b) 0 (c) $\frac{4}{12}$

7.(a) $\frac{n-3}{n-1} \frac{n-3}{n} \frac{n-4}{n-1}$ (b) Correct rearrangement with at least one further step to reach $n^2 - n - 90 = 0$ (c) 10

8.(a) $\frac{8}{15}$ (b) $\frac{168}{210}$

9.(a) $\frac{2}{6}, \frac{2}{5}, \frac{4}{5}, \frac{1}{5}$ (b) $\frac{14}{30}$

10.(a) Correct tree diagram with four branches added and the five correct probabilities $\frac{2}{3}, \frac{1}{2}, \frac{1}{2}, 1, [0]$
(b) 0